

# The Lifetime Incidence of Transfer Programs

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# Motivation

## How targeted is the U.S. safety net at those in greatest need?

- Current income measures consumption imperfectly, esp. at bottom of distribution
  - Lifetime incidence may be a more informative measure of distributional impact (Modigliani & Brumberg 1954, Friedman 1957, Meyer & Sullivan 2003)
- Mobility suggests lifetime benefits could be less progressive than current benefits
  - Due to mobility, many individuals are eligible for transfers for part of their lifetimes
- Yet both eligibility rules & take-up behavior affect incidence of transfers
  - Contrast with lifetime tax burdens (Poterba 1989/1991, Fullerton & Lim 1993)
    - Eligibility rules may screen out lifetime non-poor (e.g., through asset tests)
    - Do transfers miss the neediest because of incomplete and selective take-up?

# This Paper

- ① Estimate lifetime incidence of 8 means-tested transfer programs
  - Lifetime incidence is *as or more* progressive than current incidence
  - Added progressivity is driven by disability benefits (SSI) and housing (PH/Section 8)
  - Important counterexample: unemployment insurance
  
- ② Decompose  $\Delta$  of lifetime vs. current incidence: mobility, eligibility, take-up
  - Develop a new distributional decomposition to distinguish their contributions
  - Mobility effect makes lifetime incidence more uniform than current incidence, but it is overwhelmed by eligibility and take-up effects
  - Eligibility rules & selective take-up both improve targeting: Lifetime earnings ranks of program participants are lower than nonparticipants with same current earnings

# Literature

- **Lifetime incidence of taxes and government policies**

Davies et al. 1984, Poterba 1989/1991, Slemrod 1992, Fullerton & Lim 1993, Lyon & Schwab 1995, Bengtsson et al. 2016, Catherine & Yannelis 2020

→ first analysis of lifetime incidence of transfers with realistic heterogeneity or incomplete take-up

- **Targeting properties of transfer programs**

Akerlof 1978, Nichols & Zeckhauser 1982, Parsons 1991, Dynarski & Scott-Clayton 2006, Kleven & Kopczuk 2011, Mullainathan & Shafrir 2013, Bhargava & Manoli 2015, Alatas et al. 2016, Deshpande & Li 2019, Finkelstein & Notowidigdo 2019, Gray 2019, Stepner 2019

→ new decomposition offers a precise & generalizable meaning to claims that eligibility rules or self-selection improve or harm targeting

# Outline

- ① Data and Measurement
- ② Current and Lifetime Incidence
- ③ Decomposition
- ④ Mechanisms & Policy Implications

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# Data Overview

## Panel Survey of Income Dynamics (PSID) waves, 1997–2017

- *Earnings:*
  - Track people from ages 18 to 65 if household head or spouse
  - High-quality panel, used extensively in research on lifetime tax burdens
- *Program Participation & Benefit Levels:*
  - Can observe SNAP, Medicaid, SSI, TANF, UI, WIC, LIHEAP, public housing, Section 8
  - Less confident in participation data: will validate results using Virginia admin data

## Other Data Sources

- Simulated eligibility using PSID + program rules
- CPS: Impute cash-equivalent values of Medicaid & housing assistance

# Estimating Lifetime Earnings

We estimate a panel data model of individual earnings by Poisson PML:

$$\log E[y_{it} | x_{it}] = \alpha_i + \alpha_t + x'_{it}\beta$$

where

- $\alpha_i$ : individual FE
- $\alpha_t$ : year FE
- $x_{it}$ : time-varying covariates (age, marital status, children, education)

**Steps to measure lifetime earnings** (Solon 1992, Mazumder 2005, Haider & Solon 2006)

- 1 Shrink estimated FEs  $\{\hat{\alpha}_i\}$  by Empirical Bayes
- 2 Fill in panel by imputing missing  $x_{it}$  and using predicted values of  $y_{it}$
- 3 For households: Add up predicted earnings and equalize

## Step 1: Empirical Bayes Shrinkage of Estimated Fixed Effects

Following Morris (1983), we shrink estimated FEs towards conditional mean predicted by time-invariant covariates (Jacobs & Lefgren 2007, Chandra et al. 2016, Sorkin 2017)

$$\alpha_i^* = \frac{\sigma_i^2}{\sigma_i^2 + \tau^2} \cdot z_i' \lambda + \frac{\tau^2}{\sigma_i^2 + \tau^2} \cdot \hat{\alpha}_i$$

where

- $z_i$ : time-invariant covariates (sex, race)
- $\sigma_i^2$ : sampling variance of  $\hat{\alpha}_i$  (sample analog obtained via bootstrap)
- $\tau^2$ : variance of true FEs  $\alpha_i$

→ Procedure iteratively updates  $\hat{\lambda}, \hat{\tau}$  until convergence

## Steps 2 and 3: Fill in Panel & Equivalize Household Earnings

**Step 2.** Impute covariates & predicted earnings if not observed at ages from 18 to 65

→ Assume that  $x_{it}$  unchanged at latest observed value

$$\widehat{y}_{it}^* = \exp \left( \widehat{\alpha}_i^* + \widehat{\alpha}_{ct} + x_{it} \widehat{\beta} \right)$$

**Step 3.** For households, add up earnings and equivalize:

$$\widehat{y}_{ht}^* = f \left( \widehat{y}_{i(h)t}^* + \widehat{y}_{j(h)t}^* \right)$$

where

- $i(h)$  and  $j(h)$ : head and spouse of household  $h$
- $f(\cdot)$ : equivalence scale function (Meyer & Sullivan 2008)

[Details](#)

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# Who Benefits from Means-Tested Transfers?

We study incidence graphically by local linear regressions of outcomes on earnings ranks.

## Outcomes:

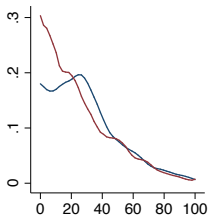
- Program participation rates
- Annual per-capita value of benefits

Throughout looking at current and lifetime earnings ranks of:

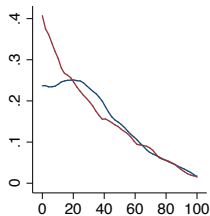
- Individuals
- Households
- Equivalized households

Probability of Program Participation

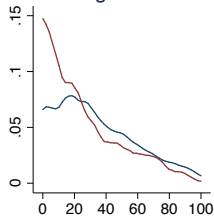
SNAP



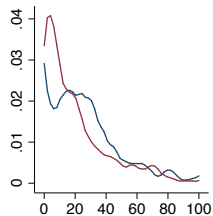
Medicaid



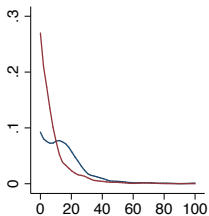
Housing Assistance



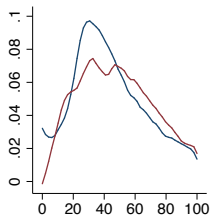
TANF



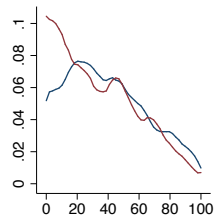
SSI



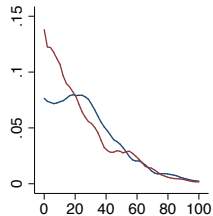
UI



WIC



LIHEAP



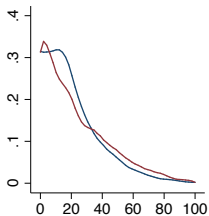
Individual Earnings Percentile

— Current

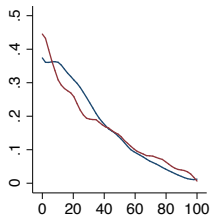
— Lifetime

Probability of Program Participation

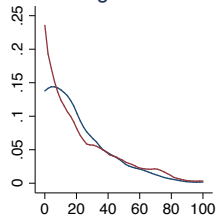
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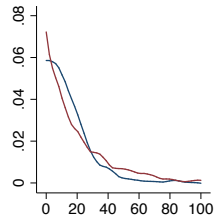
Medicaid



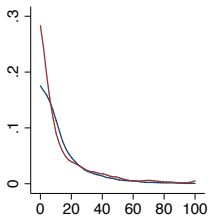
Housing Assistance



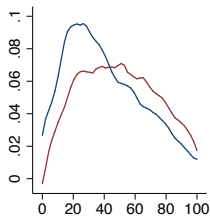
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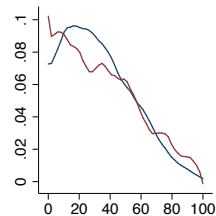
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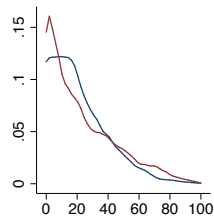
UI



WIC



LIHEAP



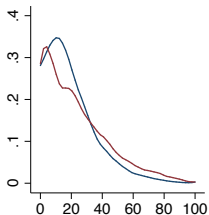
Household Earnings Percentile

— Current

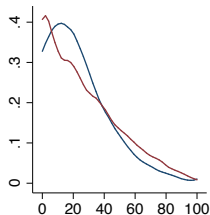
— Lifetime

Probability of Program Participation

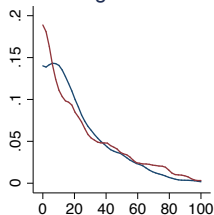
SNAP



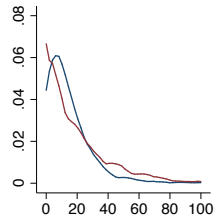
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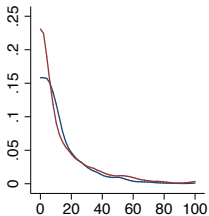
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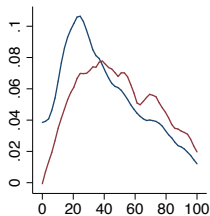
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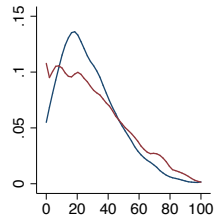
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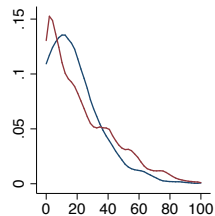
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WIC



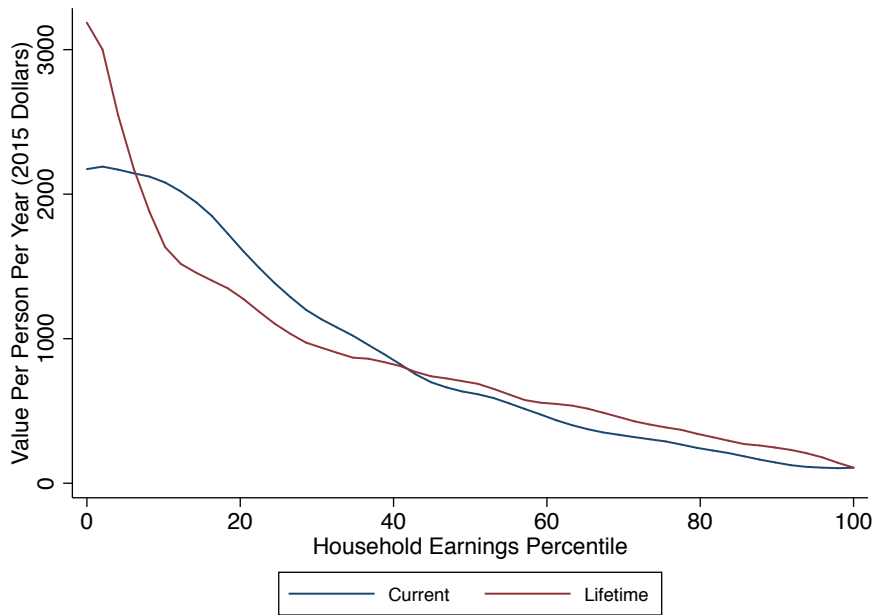
LIHEAP



Equivalized Household Earnings Percentile

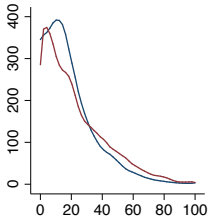
— Current

— Lifetime

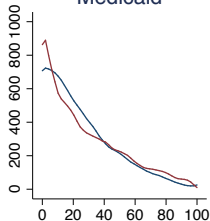


Value Per Person Per Year (2015 Dollars)

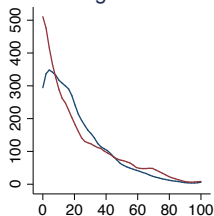
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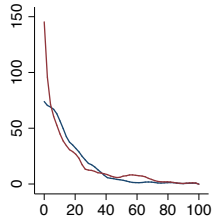
Medicaid



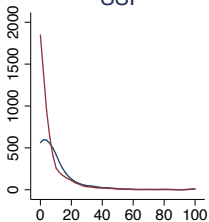
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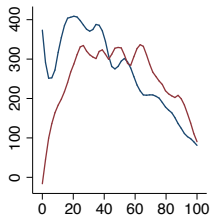
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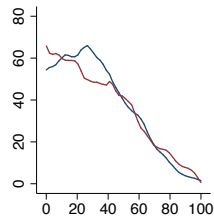
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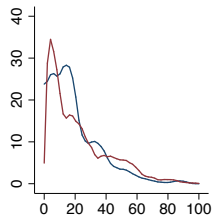
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WIC



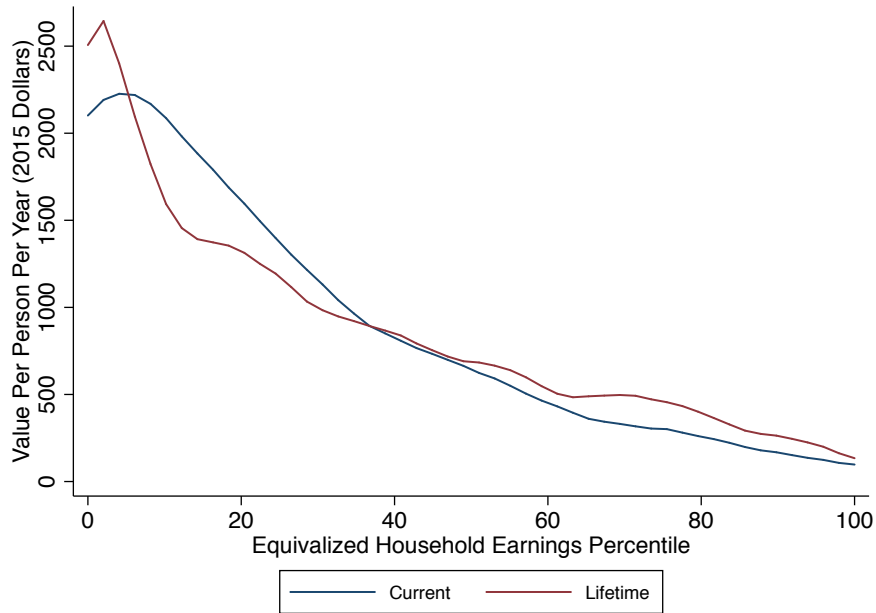
LIHEAP



Household Earnings Percentile

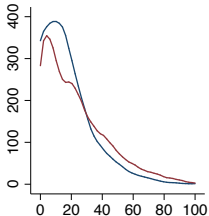
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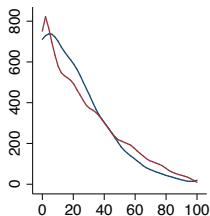


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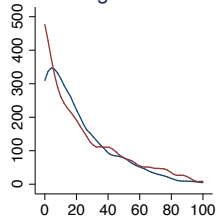
SNAP



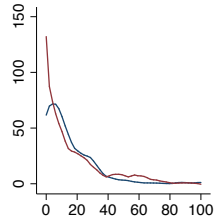
Medicaid



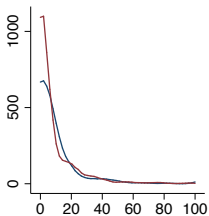
Housing Assistance



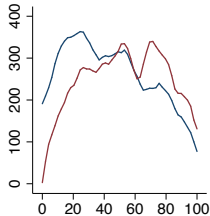
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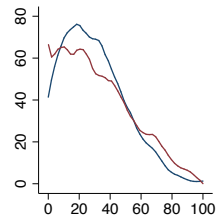
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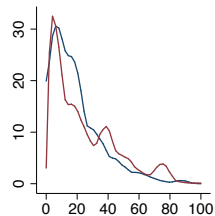
UI



WIC



LIHEAP



Equivalized Household Earnings Percentile

— Current

— Lifetime

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# Mobility, Eligibility, and Take-Up Effects

Why is lifetime incidence more or less progressive than current incidence?

3 forces to explain difference in incidence:

- ① **Mobility:** Life-cycle trends or transitory shocks cause people to change current earnings ranks over their lifetimes
- ② **Eligibility:** Screening criteria other than current earnings may have residual predictive power for lifetime earnings
- ③ **Take-Up:** Among eligibles, take-up probability depends on lifetime earnings rank, not only current earnings rank

→ Will walk through decomposition for program participation, but it is broadly applicable

## Decomposition

Let  $D_{it} = 1$  indicate participation by  $i$  in  $t$ . By the law of conditional expectation:

$$\Pr(D_{it} = 1 \mid \bar{r}_i = r) = \int \Pr(D_{it} = 1 \mid r_{it}, \bar{r}_i = r) dF(r_{it} \mid \bar{r}_i = r).$$

We use this to define two participation counterfactuals:

- **Mobility effect only:** no conditioning on  $\bar{r}_i$

$$q(r) = \int \Pr(D_{it} = 1 \mid r_{it}) dF(r_{it} \mid \bar{r}_i).$$

- **Mobility & eligibility effects:** eligibility  $E_{it}$  is informative about  $\bar{r}_i$

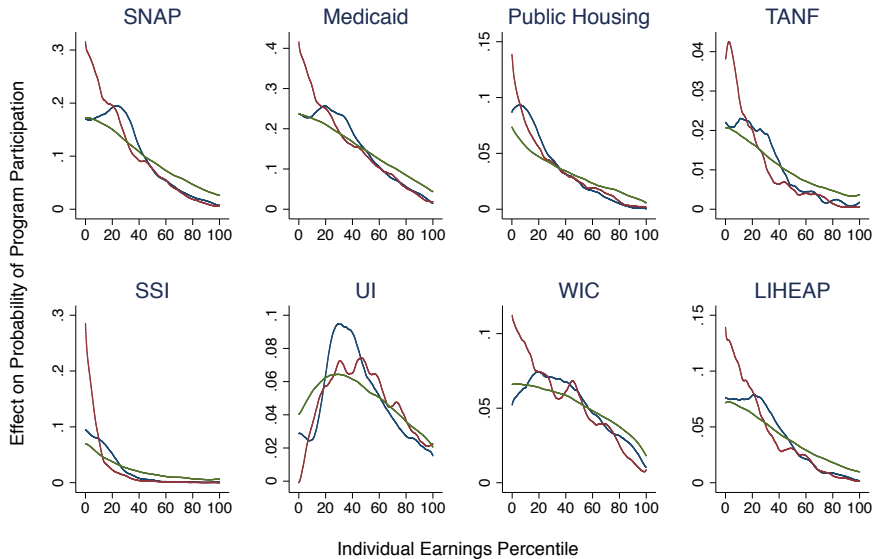
$$s(r) = \int \Pr(D_{it} = 1 \mid r_{it}, E_{it} = 1) dF(r_{it} \mid \bar{r}_i = r),$$

# Decomposition

Let  $\bar{p}(r) = \Pr(D_{it} = 1 \mid \bar{r}_i = r)$  and  $p(r) = \Pr(D_{it} = 1 \mid r_{it} = r)$ . The decomposition is:

$$\bar{p}(r) - p(r) = \underbrace{[\bar{p}(r) - s(r)]}_{\text{take-up effect}} + \underbrace{[s(r) - q(r)]}_{\text{eligibility effect}} + \underbrace{[q(r) - p(r)]}_{\text{mobility effect}}.$$

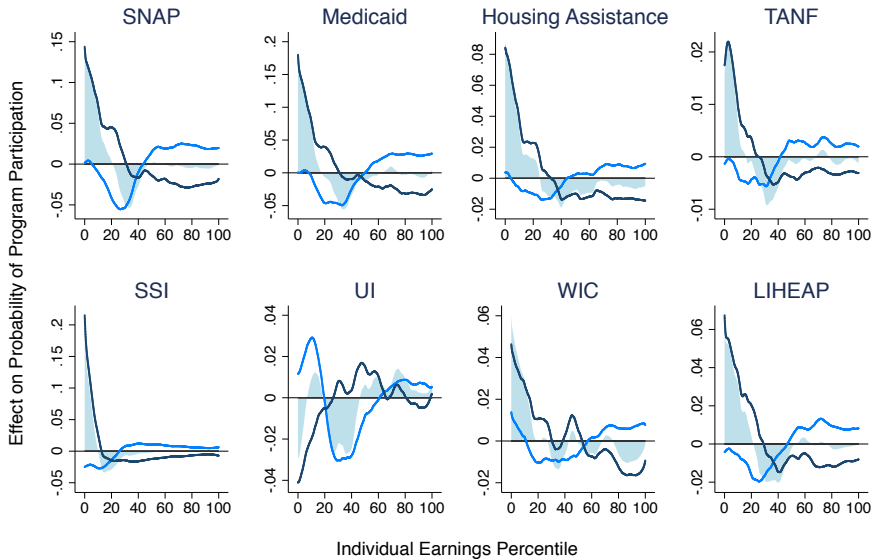
- **Mobility Effect:** Current incidence vs. lifetime incidence in counterfactual where participation is uninformative about lifetime earnings given current earnings
- **Eligibility Effect:** Comparison of lifetime incidence in two counterfactuals: participation is uninformative vs. participation is informative only via eligibility
- **Take-Up Effect:** Actual lifetime incidence vs. lifetime incidence in counterfactual in which participation is informative only via eligibility

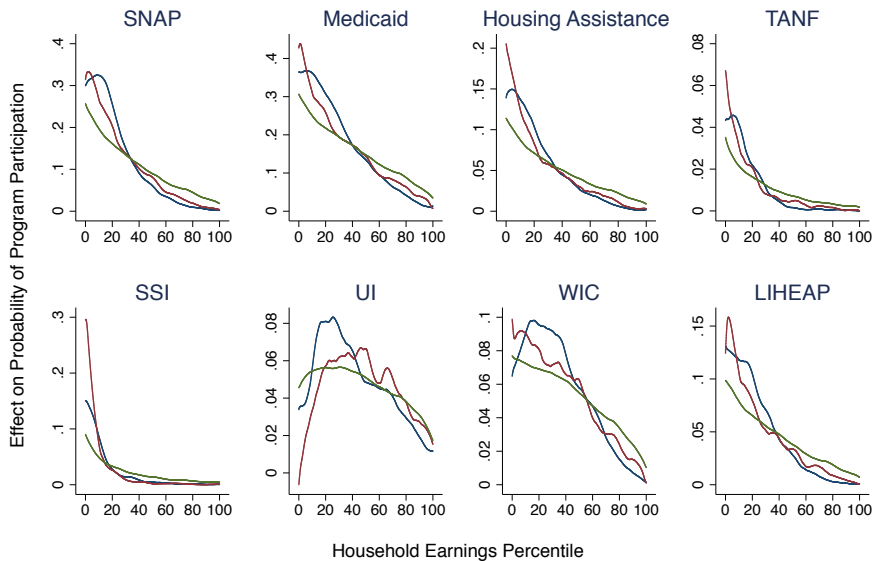


— Current, Actual

— Lifetime, Actual

— Lifetime, Counterfactual

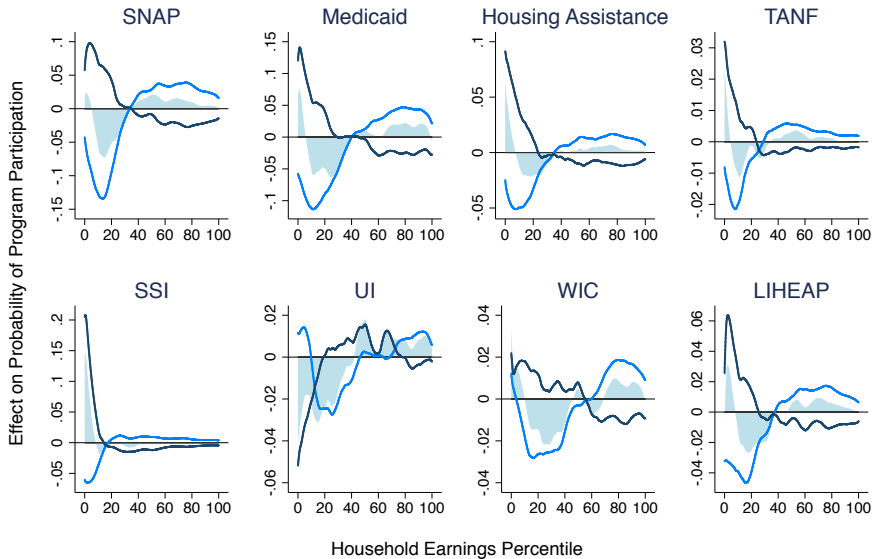


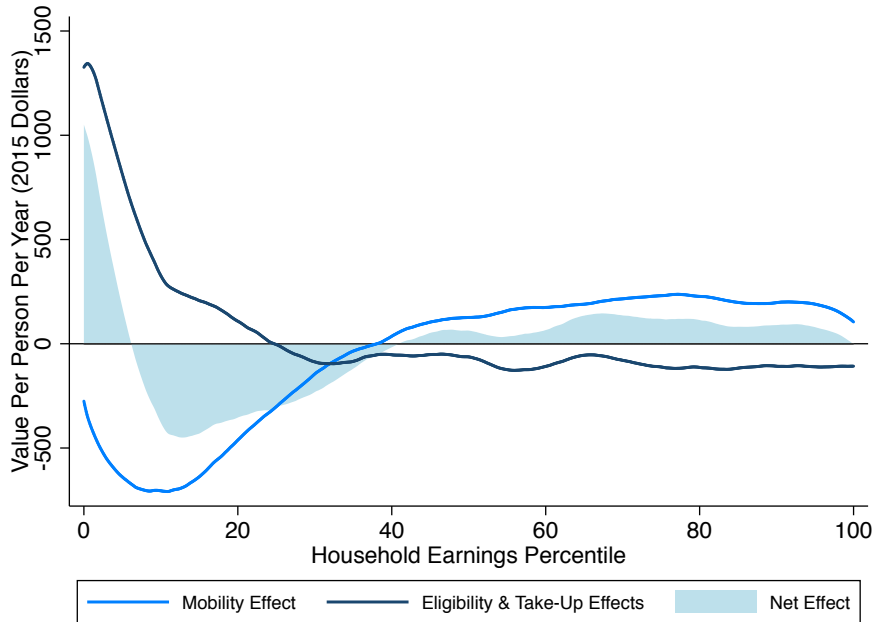


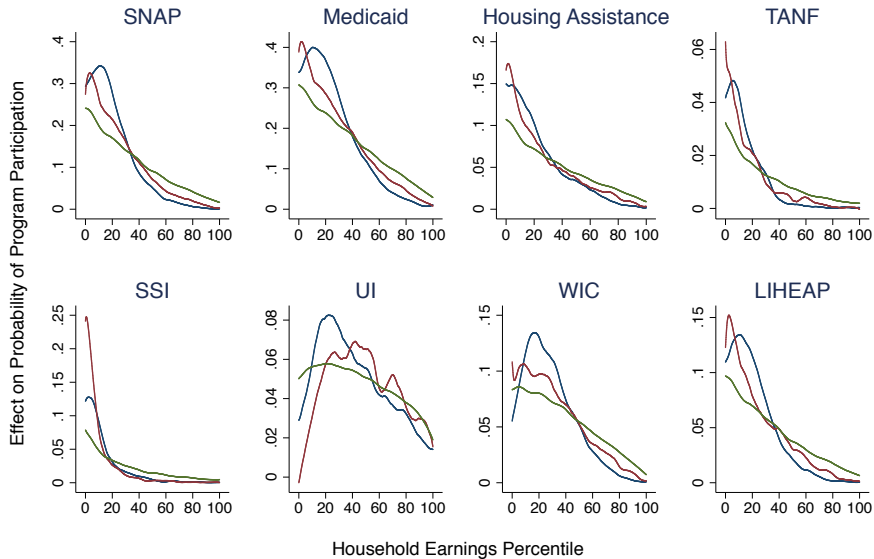
— Current, Actual

— Lifetime, Actual

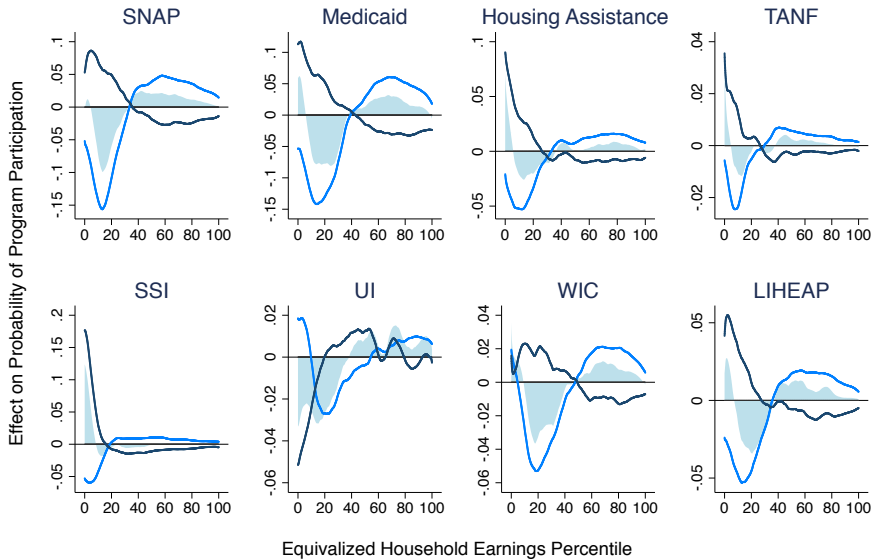
— Lifetime, Counterfactual







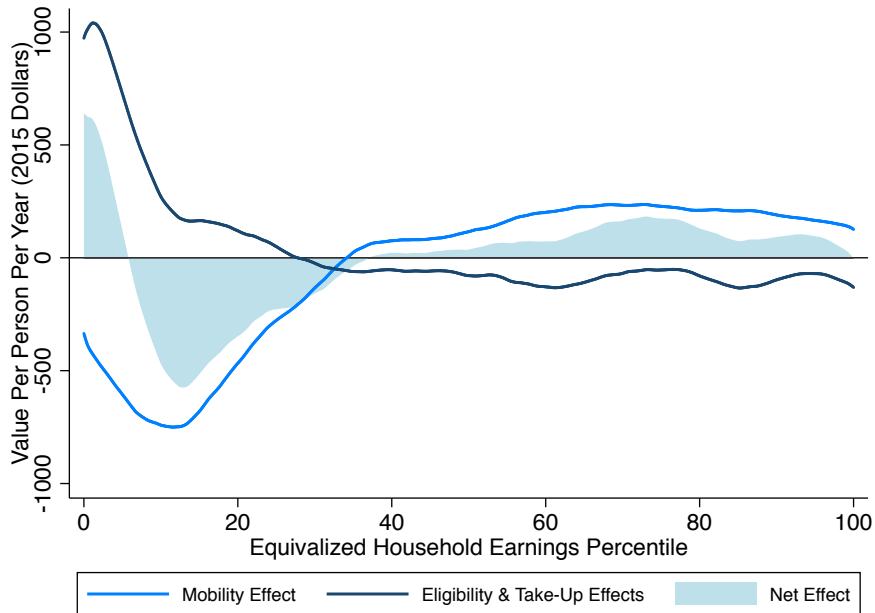
— Current, Actual — Lifetime, Actual — Lifetime, Counterfactual



— Mobility Effect

— Eligibility & Take-Up Effects

Net Effect

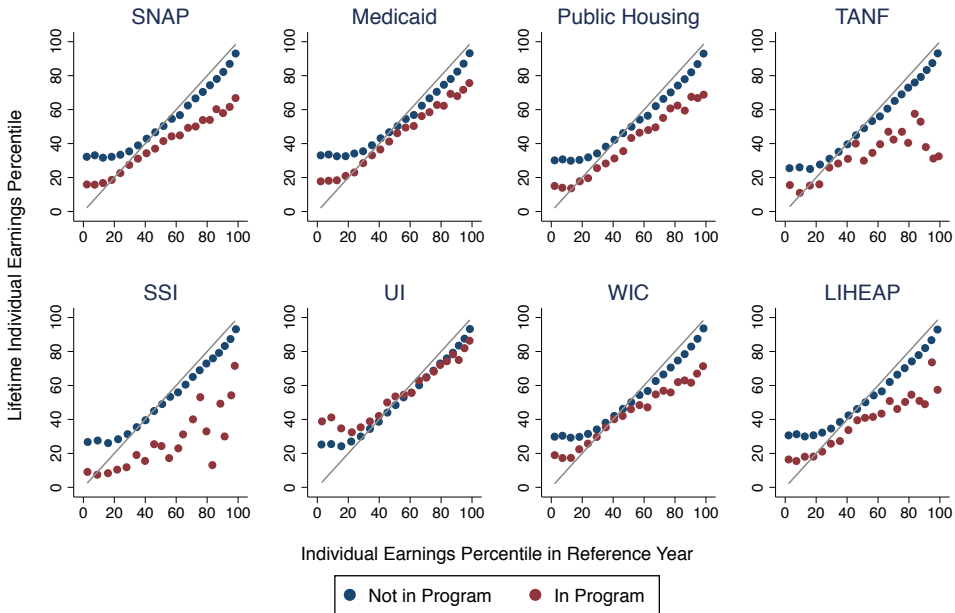


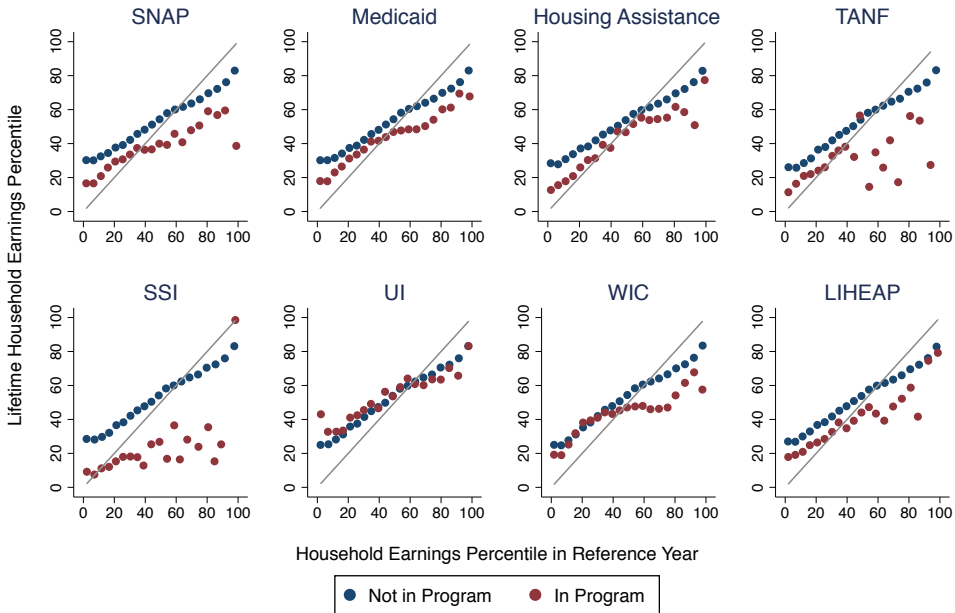
# Outline

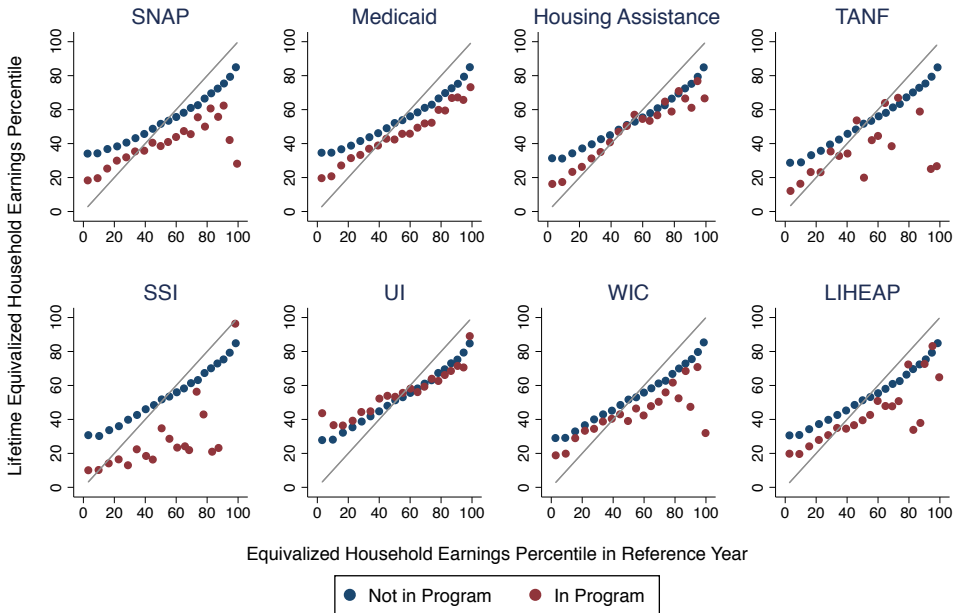
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# Mechanisms

- So far only indirect evidence for eligibility & take-up channels: actual and counterfactual lifetime incidence are highly dissimilar
- Now, direct evidence: Lifetime earnings ranks of program participants are consistently lower than nonparticipants at the same earnings rank
- To do: Distinguish between nonparticipant eligible and nonparticipant ineligible







# Policy Implications

Related to famous case for lifetime income taxation:

- Vickrey (1939, 1946): “one of the principal defects” of income tax system is the nonlinear taxation of current, instead of lifetime, income
- Few means of lifetime redistribution in tax code, other than Social Security benefit formula (Diamond 2003, Kaplow 2007)
- Our argument: Means-tested transfers replicate a nonlinear lifetime income tax at the bottom, redistributing to lifetime poor to an extent difficult/distortionary to replicate via nonlinear current income taxation
- Suggests a substantial efficiency benefit from “tagging” via eligibility rules and selective take-up

# Next Steps

## Suggestions?

- Code up eligibility rules & implement full decomposition
  - Decompose mobility into lifecycle & shock components
  - Understanding take-up: driven by observable predictors of lifetime earnings?
- Rank households on current consumption
  - Impute consumption as in Blundell et al. (2008)
  - Present direct evidence on consumption from quality of durable goods (e.g., car value)
- Estimate insurance value of safety net & account for in valuation of benefits

# Equivalence Scale Details

$$y_{ht} = \frac{1}{\psi_h} \sum_{i \in h} y_{it}$$

Options:

- Two-parameter family

$$\psi_h = (N_h^{\text{adults}} + \beta \cdot N_h^{\text{children}})^{\gamma}$$

$$\beta = 0.7, \gamma = 0.7$$

- Square-root

$$\psi_h = (N_h^{\text{adults}} + N_h^{\text{children}})^{1/2}$$

- OECD

$$\psi_h = \left( 1 + 0.5 \cdot (N_h^{\text{adults}} - 1) + 0.3 \cdot N_h^{\text{children}} \right)$$

# Earnings Percentile Rank Definitions

Individual current earnings rank

$$r_{it} = \Pr(y \leq y_{it})$$

Household current earnings rank

$$r_{ht} = \Pr(y \leq y_{ht})$$

Individual lifetime earnings rank

$$\bar{r}_i = \Pr\left(y \leq T^{-1} \sum_t \hat{y}_{it}^*\right)$$

Household lifetime earnings rank

$$\bar{r}_h = \Pr\left(y \leq T^{-1} \sum_t \hat{y}_{ht}^*\right)$$